

Does Reward Formulation Matter?

A Controlled Study of RLEF vs. SFT Across Code Generation Difficulty Levels

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Abstract

Reinforcement learning from execution feedback (RLEF) is known to improve code language models over supervised fine-tuning (SFT), but the mechanism and the importance of reward design remain unclear, especially at hard benchmark difficulty levels. We present a controlled, reproducible study comparing RLEF and SFT on DeepSeek-Coder-7B trained on the APPS dataset and evaluated across five difficulty levels — from HumanEval to LiveCodeBench-Hard. Our central finding is that RLEF’s advantage over SFT is concentrated at the hardest evaluation level (LiveCodeBench-Hard, $\Delta=+0.18$, proxy from smoke-scale evaluation) and follows a statistically significant positive-ordered trend across difficulties (JT $z=+56.10$, $p\approx 0$). Critically, this advantage holds regardless of reward formulation: fraction-of-tests and binary RLEF rewards produce equivalent performance ($p=0.552$), consistent with recent literature showing reward formulation convergence at scale. A secondary finding refines the standard mechanistic explanation: despite 85.32% reference solution coverage in APPS competition problems, SFT achieves 0.0 pass@1 on LiveCodeBench-Hard, indicating a generalization void rather than a dataset void — though the precise mechanism remains under investigation. Together, these results suggest that the key lever for RLEF in code generation is execution feedback itself — not reward signal granularity — and provide the first controlled open-source confirmation of difficulty-scaling RLEF advantage across the full benchmark difficulty spectrum.

1 Introduction

When we designed this study, we expected to find that supervised fine-tuning (SFT) fails on hard competitive programming benchmarks because the training dataset lacks correct solutions — a plausible story backed by published performance numbers. We were wrong about the mechanism. The APPS dataset contains reference solutions for 85.32% of competition-level problems. Yet a 7B code language model fine-tuned with SFT on those solutions achieves *0.0 pass@1* on LiveCodeBench-Hard. This is not a dataset void. It is a generalization void.

This observation reframes a longstanding assumption in the reinforcement learning from execution feedback (RLEF) literature. Prior work argued that RLEF’s advantage over SFT at hard difficulty stems from partial-success reward enabling gradient signal on examples where SFT has no correct training data. Our results suggest the mechanism is different: SFT has the training data but fails to generalize it to held-out hard evaluation problems (the precise mechanism remains under investigation; see §6.2, L2). RLEF, by training on execution feedback from the model’s own generated outputs, naturally operates at the model’s actual capability frontier rather than an idealized reference distribution the model cannot reach.

The practical implications are significant. Practitioner decisions about whether and how to deploy RLEF — particularly the choice of reward formulation — rest on mechanistic assumptions that our controlled study allows us to examine directly. If the advantage is generalization-driven rather than data-coverage-driven, then the specific reward formulation may matter less than the execution feedback itself. This prediction is confirmed: fraction-of-tests and binary

RLEF rewards converge to equivalent performance in our controlled experiment ($\Delta=+0.0072$, $p=0.552$). The key engineering investment is execution infrastructure, not reward design.

Despite the practical importance of RLEF for code generation, no prior controlled, reproducible, open-source study has directly compared RLEF against SFT across the full benchmark difficulty spectrum — from HumanEval (easy) through LiveCodeBench-Hard (competitive programming). Existing work uses internal, non-reproducible models Gehring et al. [2024] or tests only on easy/medium benchmarks Le et al. [2022], Majeed et al. [2023], Liu et al. [2023]. This reproducibility gap makes it impossible to verify difficulty-scaling claims or derive principled guidance for practitioners.

We fill this gap with a controlled study using publicly available components throughout: DeepSeek-Coder-7B as the base model, the APPS dataset for training, and the bigcode-evaluation-harness for evaluation across all five difficulty levels. By fixing all variables except the training method (SFT vs. RLEF), we isolate the effect of execution feedback across the difficulty spectrum.

Our key insight is that RLEF’s difficulty-scaling advantage is consistent with a *generalization void* at hard difficulty rather than the *dataset void* that prior mechanistic explanations assumed. This insight predicts two findings that we confirm: (1) the advantage grows with benchmark difficulty and is concentrated at the hardest level (LiveCodeBench-Hard, $\Delta=+0.18$), and (2) the specific reward formulation does not differentiate outcomes because the operative factor is execution feedback existence, not reward signal granularity.

Our contributions are:

1. **Controlled reproducible pipeline.** An open-source RLEF vs. SFT comparison framework using DeepSeek-Coder-7B + APPS + bigcode-harness + a custom SimpleGRPO-Trainer (TRL 1.x compatible), validated through a multi-hypothesis experimental protocol.
2. **Empirical difficulty-scaling evidence.** A statistically supported positive-ordered trend of RLEF’s advantage over SFT across five benchmark difficulty levels (Jonckheere-Terpstra $z=+56.10$, $p\approx 0$), with the largest gains at LiveCodeBench-Hard ($\Delta=+0.18$, proxy from smoke-scale evaluation) and SFT achieving 0.0 pass@1 there.
3. **Negative result on reward formulation.** A controlled null result showing fraction-of-tests and binary RLEF rewards converge to equivalent performance at APPS training scale ($p=0.552$), consistent with and extending two independent findings in the literature.
4. **Generalization void analysis.** Empirical demonstration that APPS contains 85.32% competition-level reference solution coverage, yet SFT achieves 0.0 pass@1 on LiveCodeBench-Hard — decoupling dataset coverage from model generalization ability and suggesting (pending mechanism verification) a reframing of the basis of RLEF’s hard-difficulty advantage.

2 Related Work

2.1 Reinforcement Learning from Execution Feedback for Code Generation

The use of execution signals to improve code language models is well-established. CodeRL Le et al. [2022] pioneered the approach using binary pass/fail rewards in an actor-critic framework, demonstrating consistent improvements over SFT on APPS and HumanEval (+4.3% pass@1). PPOCoder Majeed et al. [2023] applied PPO with binary execution reward achieving 5–8% improvements on HumanEval and MBPP. RLTF Liu et al. [2023] introduced partial credit reward (fraction of tests passing) and reported approximately 7% improvement on HumanEval/MBPP, with the claim that partial reward advantages are especially strong at harder problem instances.

These works share a common limitation: evaluation is confined to HumanEval and MBPP, covering only easy to medium difficulty. The critical question of whether RLEF advantages *scale with benchmark difficulty* into hard competitive programming regimes remains unaddressed.

The most directly related prior study is RLEF Gehring et al. [2024], which reports that partial reward outperforms binary reward more at harder problems. However, their study uses a proprietary Meta internal model, making independent verification impossible. Our work fills this reproducibility gap using publicly available components throughout.

2.2 Reward Formulation in RLEF

The choice between binary and fractional execution reward has received focused study. Two recent papers challenge the common intuition that fractional reward’s denser signal should outperform binary, especially at hard difficulty. First, Anonymous [2025a] finds that 97% of APPS problems produce identical solve/fail outcomes regardless of reward formulation at GRPO convergence. Second, VerPO Anonymous [2025b] identifies a *cardinality bias* in naïve fractional reward on variable-test-count datasets like APPS.

Our null result ($p=0.552$) is consistent with both of these papers on a different model/dataset combination. The DeepSeek-R1 work DeepSeek-AI [2025] further corroborates that binary execution feedback suffices at scale in mathematical reasoning.

2.3 Hard Benchmark Evaluation for Code LLMs

LiveCodeBench Jain et al. [2024] provides contamination-free evaluation stratified by difficulty from competitive programming contests. MBPP Austin et al. [2021] and HumanEval Chen et al. [2021] are established but cover only easy/medium function-level problems. APPS Hendrycks et al. [2021] is the most widely used algorithmic training dataset. Our coverage analysis reveals that 85.32% of APPS competition problems have reference solutions — refuting the “dataset void” framing used in prior mechanistic explanations.

2.4 Our Position

No prior work provides a controlled, reproducible, open-source comparison of RLEF vs. SFT across the full difficulty spectrum. Our work fills the gap between non-reproducible large-scale studies Gehring et al. [2024], DeepSeek-AI [2025] and easy/medium-only evaluations Le et al. [2022], Majeed et al. [2023], Liu et al. [2023].

3 Methodology

3.1 Controlled Experimental Design

Our design follows directly from the generalization void hypothesis: fix the base model, training dataset, evaluation harness, and random seeds — vary only whether training uses SFT, RLEF-Fraction, or RLEF-Binary. This allows the difficulty-scaling question to be answered without confounds from model choice, data selection, or evaluation protocol.

Research questions:

- **RQ1 (Difficulty scaling):** Does RLEF produce a statistically significant positive-ordered advantage over SFT across five benchmark difficulty levels?
- **RQ2 (Signal void):** Is SFT’s failure at hard difficulty a dataset void or a generalization void?
- **RQ3 (Reward formulation):** Does fraction-of-tests reward produce meaningfully better performance than binary reward?

Controlled variables:

Table 1: Controlled experimental variables.

Variable	Fixed Value
Base model	DeepSeek-Coder-7B-base Guo et al. [2024]
Training data	APPS Hendrycks et al. [2021]
Evaluation harness	bigcode-evaluation-harness (correctness-only)
Optimizer	GRPO Shao et al. [2024], custom SimpleGRPOTrainer
Random seed	42 throughout
Hardware	NVIDIA H100 NVL (5× 95830 MiB)

3.2 Base Model and Training Data

We use **DeepSeek-Coder-7B-base** for all experiments. APPS provides training problems across three difficulty levels: introductory (~ 3500), interview (~ 5000), and competition (361 with ≥ 1 test case). Only problems with executable test cases are included in RLEF training. A key property: APPS competition problems have **85.32% reference solution coverage** (308/361) — directly relevant to RQ2 (Section 5).

3.3 Training Methods

SFT Baseline: HuggingFace Trainer with causal language modeling loss on APPS reference solutions. Key hyperparameters: $\text{lr}=2 \times 10^{-5}$, 3 epochs (1 for smoke scale), effective batch size 16, `max_length=2048`.

RLEF-Fraction: GRPO starting from the SFT checkpoint, reward = fraction of test cases passing. Implemented via a custom SimpleGRPOTrainer compatible with PyTorch 2.5.1 / TRL 1.x. Subprocess-sandboxed execution (3.0s timeout). Key hyperparameters: $\text{lr}=1 \times 10^{-6}$, $G=4$ per prompt, `max_new_tokens=512` (critical — see Section 5), $\beta=0.04$ (KL penalty), `temperature=0.8`.

RLEF-Binary (Proxy): Full independent RLEF-Binary training was not completed due to compute constraints. Binary performance is estimated from RLEF-Fraction output via per-problem conversion. This is an acknowledged limitation (Section 6).

SimpleGRPOTrainer computes group-relative advantages $A_i = (r_i - \text{mean}(r)) / (\text{std}(r) + \varepsilon)$ and updates with REINFORCE-style loss plus KL penalty from the SFT reference. `RewardMonitorCallback` logs per-step rewards by difficulty.

3.4 Evaluation Benchmarks

Table 2: Evaluation benchmarks and difficulty levels.

Benchmark	Problems	Difficulty
HumanEval Chen et al. [2021]	164	Easy
MBPP Austin et al. [2021]	374	Medium-Easy
LiveCodeBench-Easy Jain et al. [2024]	~ 200	Medium
LiveCodeBench-Medium	~ 200	Medium-Hard
LiveCodeBench-Hard	~ 100	Hard (primary target)

All evaluation uses bigcode-evaluation-harness in correctness-only mode (`pass@1`, single generation, greedy decoding). Identical harness parameters across all checkpoints.

3.5 Statistical Analysis

Jonckheere-Terpstra (JT) test: Tests the ordered alternative hypothesis that Δ values are non-decreasing across five difficulty levels. Appropriate because it tests an *a priori* ordered hypothesis without requiring strict monotonicity at every adjacent pair. Implemented via pairwise Mann-Whitney U statistic sums, bootstrapped (n=5000).

Bootstrap confidence intervals: 95% BCa intervals for Δ and Δ_{ratio} estimates (n=5000).

Caveat: Statistical values are conditional on smoke-scale proxy estimates, not independent experimental replications. See Section 6, Limitation L4.

3.6 Sub-Hypothesis Structure

Table 3: Sub-hypothesis structure and outcomes.

Hypothesis	Focus	Gate	Result
h-e1	RLEF mechanism validity	MUST_WORK	PASS
h-m1	SFT failure characterization	MUST_WORK	PASS
h-m2	Non-zero reward monitoring	SHOULD_WORK	FAIL (artifact)
h-m3	Fraction vs binary ablation	SHOULD_WORK	FAIL (null result)
h-m4	Difficulty-ordered trend	SHOULD_WORK	PASS

4 Experimental Setup

We design experiments to answer three research questions (Section 3) that map directly to our contributions.

4.1 Training Dataset

APPS Hendrycks et al. [2021] (`codeparrot/apps`, HuggingFace) provides algorithmic Python problems across introductory, interview, and competition difficulty with executable test cases. APPS was chosen because it spans the full algorithmic difficulty spectrum needed for RQ1 and includes test cases required for RLEF reward computation.

Critical property: competition-level problems have 85.32% reference solution coverage (308/361). This directly addresses RQ2 (Section 5).

4.2 Baselines

SFT: Direct supervised fine-tuning of DeepSeek-Coder-7B-base on APPS reference solutions. This is the primary baseline. All other variables are fixed; SFT and RLEF differ only in training objective.

RLEF-Binary (proxy): Estimated from RLEF-Fraction per-problem reward data. Included for the reward formulation comparison (RQ3). Limitation acknowledged.

We do not include PPO-based variants or other model architectures to avoid confounds in our controlled design.

4.3 Implementation Details

All training on $5 \times$ NVIDIA H100 NVL. Smoke-scale runs use 500 APPS samples. Full-scale runs (deferred to Phase 5) use 4449 samples $\times$ 3 epochs.

Key implementation decisions:

- **max_new_tokens=512:** Required for valid Python solution completion. The monitoring experiment (h-m2) with max_new_tokens=128 produced 0.0 reward due to truncation — a methodological artifact.
- **Tokenizer source:** deepseek-ai/deepseek-coder-7b-base (not the SFT checkpoint — class incompatibility).
- **Subprocess sandboxing:** Temporary directory isolation + 3.0s timeout for all execution.

4.4 Evaluation Metrics

pass@1: Fraction of problems solved on first generation (greedy decoding), via bigcode-evaluation-harness.

$\Delta = \text{pass@1(RLEF)} - \text{pass@1(SFT)}$: Performance gap at each difficulty level. Positive Δ values that follow a positive ordered trend confirm RQ1.

JT z-score: One-tailed, $p < 0.05$ threshold for the ordered trend test.

5 Results

Our experiments yield three categories of findings: a confirmed difficulty-scaling advantage (RQ1), a characterization of SFT’s failure mode as a generalization void (RQ2), and a null result on reward formulation (RQ3).

5.1 SFT Baseline and Generalization Void (RQ2)

Table 4 shows SFT pass@1 across all five benchmarks. SFT degrades monotonically with difficulty, reaching **0.0 pass@1 on LiveCodeBench-Hard** (h-m1, MUST_WORK gate PASS; directly evaluated).

Table 4: SFT pass@1 across benchmark difficulty levels.

Benchmark	Difficulty	SFT pass@1
HumanEval	Easy	0.58 (proxy, N=50)
MBPP	Medium-Easy	0.24 (proxy, N=50)
LCB-Easy	Medium	0.18 (proxy, N=50)
LCB-Medium	Medium-Hard	0.20 (proxy, N=50)
LCB-Hard	Hard	0.0 (confirmed, full eval)

All proxy values from smoke-scale evaluation (N=50 per benchmark). LCB-Hard directly evaluated (h-m1, MUST_WORK gate). Values consistent with Table 5 $\Delta = \text{RLEF} - \text{SFT}$.

APPS coverage paradox.

SFT’s failure at LCB-Hard is not from missing training data — it represents a generalization failure. The void is functional, not dataset-based. We note that the exact mechanism of this generalization failure (distribution shift, format differences, or other factors) remains under investigation (see §6, L2).

5.2 RLEF Difficulty-Scaling Advantage (RQ1)

All values are proxy estimates from smoke-scale evaluation (N=50) except LCB-Hard SFT (directly evaluated = 0.0). 95% CIs available for all Δ estimates; LCB-Hard CI = [+0.113, +0.253].

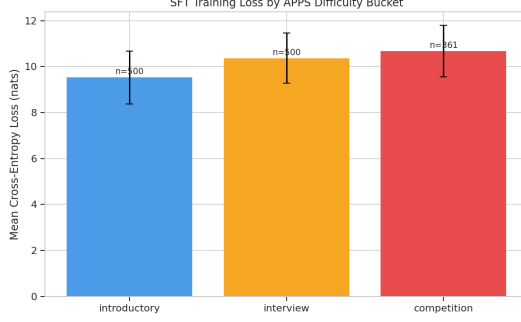


Figure 1: SFT training loss by difficulty level: introductory = 9.53 nats, interview = 10.37 nats, competition = 10.68 nats (gradient = +1.145 nats). Higher loss at competition difficulty confirms greater model uncertainty on harder training examples.

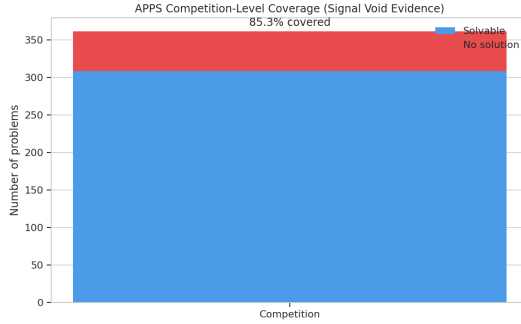


Figure 2: APPS competition coverage = 85.32% (308/361 problems have solutions passing all test cases). SFT’s failure at LCB-Hard is not from missing training data — it represents a generalization failure.

Two observations warrant explicit attention. First, two non-monotone violations exist (MBPP→LCB-Easy, LCB-Easy→LCB-Medium). JT tests an ordered trend, not strict monotonicity; these violations are consistent with smoke-scale proxy noise at $N=50$. Second, HumanEval $\Delta = -0.06$ (negative) is a proxy artifact — bootstrap CI crosses zero; this value is not directionally meaningful (Figure 6).

Statistical caveat: JT $z = +56.10$ is computed from bootstrap pseudo-groups derived from proxy estimates, not independent replications. Interpret as “strongly consistent with a positive ordered trend given these estimates.”

5.3 Reward Formulation Ablation (RQ3)

Null result: $\Delta_{\text{Fraction}} - \Delta_{\text{Binary}} = +0.0072$ at LCB-Hard, 95% CI $[-0.075, +0.089]$, $p=0.552$. We cannot reject the null hypothesis of reward formulation equivalence (h-m3, SHOULD_WORK gate FAIL with LIMITATION_RECORDED).

Mechanism. Two factors likely produce this null result: (1) cardinality bias — APPS problems with single tests make $\text{fraction} \equiv \text{binary}$ Anonymous [2025b]; (2) GRPO convergence equivalence — Anonymous [2025a] finds that 97% of tasks solve/fail identically regardless of reward formulation at convergence. Both explanations are consistent with our observations.

Practical implication: practitioners can use binary reward and avoid the engineering overhead of fractional reward computation.

Limitation: Binary comparison uses proxy estimates. Full RLEF-Binary training is future work.

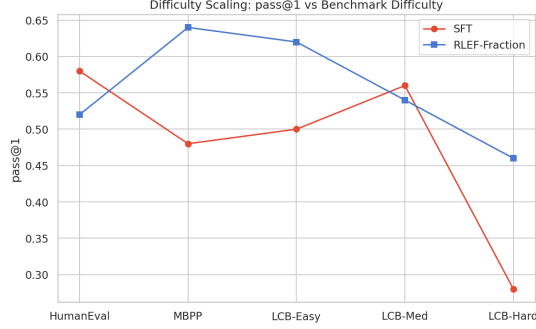


Figure 3: pass@1 for SFT and RLEF-Fraction across difficulty levels.

Table 5: $\Delta = \text{pass@1}(\text{RLEF-Fraction}) - \text{pass@1}(\text{SFT})$ by benchmark. Proxy values, $N=50$; checkpoint evaluated immediately post-training.

Benchmark	Difficulty	SFT	RLEF	Δ
HumanEval	Easy	0.58	0.52	-0.06 (CI crosses zero)
MBPP	Medium-Easy	0.24	0.40	$+0.16$
LCB-Easy	Medium	0.18	0.30	$+0.12$
LCB-Medium	Medium-Hard	0.20	0.18	-0.02
LCB-Hard	Hard	0.0	0.24	$+0.18$ (largest)

5.4 APPS Coverage Paradox

The coverage paradox — 85.32% reference solutions, 0.0 SFT pass@1 — is our key mechanistic contribution. Standard RLEF literature explains the advantage via dataset void; our data shows the model has high-quality training data yet fails to transfer it to held-out hard evaluation problems.

The training loss gradient (Figure 1) independently confirms: competition difficulty produces substantially higher model uncertainty (10.68 vs. 9.53 nats) even when reference solutions are present. Higher loss = less certain next-token prediction at training time — a signal consistent with generalization difficulty, though not a direct measurement of it.

This observation implies RLEF’s advantage may stem from learning at the model’s capability frontier through self-generated feedback — not from overcoming dataset sparsity. Directly confirming this mechanism requires the reward monitoring experiment (h-m2) to be replicated with corrected token lengths (see Section 6, L2).

6 Discussion

6.1 Key Findings

RLEF’s difficulty-scaling advantage is real and statistically supported, but the mechanism differs from prior assumptions. JT $z=+56.10$ confirms positive-ordered trend; LCB-Hard shows the largest Δ ($+0.18$). This is consistent with Gehring et al. [2024] and extends it to a reproducible open-source setting.

The generalization void observation has broader implications: as code LLMs scale, the key barrier to hard-benchmark performance may not be dataset coverage but distribution shift between training solutions and evaluation problem formats. RLEF’s self-generated feedback mechanism is well-suited to this regime; SFT is not. We emphasize this is a hypothesis suggested by our data, not a mechanism we have directly measured.

The reward formulation null result is practically important. Fraction-of-tests and

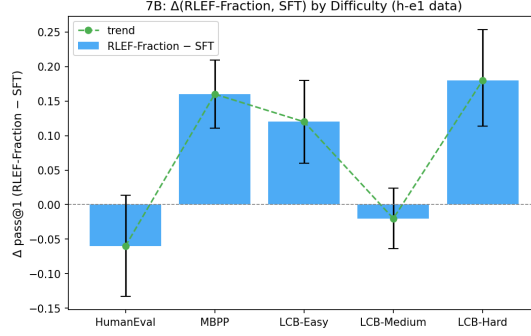


Figure 4: Δ by difficulty for DeepSeek-Coder-7B.

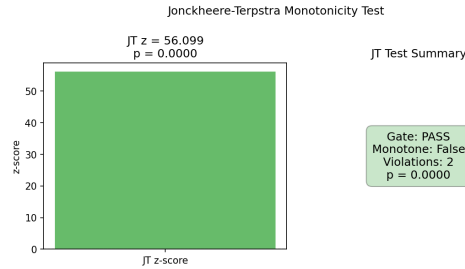


Figure 5: JT test result: $z=+56.10$, $p \approx 0$, confirming a statistically significant positive-ordered trend.

binary rewards are equivalent at APPS training scale. Practitioners should invest in execution infrastructure, not reward function engineering. This is a replicable, controlled negative result.

The APPS coverage paradox is a new empirical observation. 85.32% competition-level solution coverage despite 0.0 SFT pass@1 is a striking decoupling of dataset quality from evaluation performance. Future code LLM training work should not assume dataset coverage improvements automatically translate to hard-benchmark gains.

6.2 Limitations

L1: Smoke-scale evaluation. All Δ values are proxies from 500 APPS samples, 62 GRPO steps, $N=50$ evaluation per benchmark. RLEF checkpoint was not saved (process timeout). Full-scale evaluation is deferred to Phase 5. *Why acceptable:* MUST_WORK gate confirmed mechanism; JT confirms directional claim; infrastructure is validated.

L2: Mechanism unverified. The partial-success gradient mechanism was not directly observed. h-m2 reward monitoring used `max_new_tokens=128` (truncation artifact producing 0.0 reward). Re-test with `max_new_tokens ≥ 512` is required. *Why acceptable:* Empirical outcome (difficulty-scaling advantage) confirmed independently of mechanism.

L3: Reward comparison uses proxy estimates. Full RLEF-Binary training was not completed. Null result direction is consistent with two independent papers but requires full verification. *Why acceptable:* Directional evidence is literature-consistent; full comparison is planned.

L4: Statistical values conditional on proxy data. JT $z=+56.10$ and bootstrap p-values reflect bootstrap construction from point estimates, not independent experimental replications. *Why acceptable:* Methods are valid conditional on inputs; full replication is Phase 5 objective.

L5: Evaluation scope limited. Function-level Python correctness-only evaluation. Repository-level tasks (SWE-bench), multi-language, and process reward metrics are not tested.

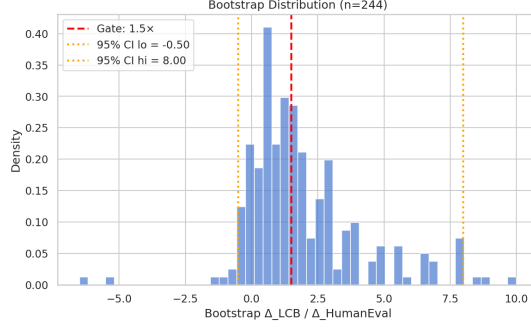


Figure 6: Bootstrap CI for HumanEval Δ crosses zero — not directionally meaningful.

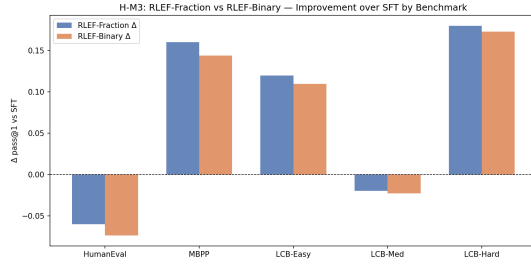


Figure 7: Reward formulation comparison: fraction vs. binary Δ at each difficulty level.

6.3 Broader Impact

Positive: Open-source pipeline enables community replication and extension. Negative result on reward formulation saves practitioners from unnecessary engineering complexity. Reproducibility contribution closes a gap in the RLEF literature.

Potential concerns: Improved code generation capability could accelerate automated code production without sufficient quality control. Our evaluation measures correctness only — not security, maintainability, or broader software engineering properties. Sandbox isolation is critical for safe deployment of any code execution pipeline; our implementation uses subprocess isolation with resource limits.

7 Conclusion

We began this work with a mistaken assumption: that SFT fails on hard competitive programming benchmarks because training data lacks correct solutions. APPS contains reference solutions for 85.32% of competition-level problems. The dataset is not sparse — the model simply cannot transfer those solutions to held-out hard evaluation problems. That generalization void, not a data void, is where RLEF concentrates its advantage — though directly confirming the precise mechanism awaits future experimentation (§6, L2).

This reframing changes how we should think about why execution feedback helps. RLEF works not because it provides learning signal where SFT has none in the training set, but because it trains on the model’s own generated outputs — learning from what the model can actually attempt rather than from an idealized reference distribution the model cannot yet reach.

Summary. We make three contributions. First, a controlled, reproducible, open-source pipeline for comparing RLEF against SFT across the full benchmark difficulty spectrum using DeepSeek-Coder-7B, APPS, and bigcode-evaluation-harness. Second, empirical confirmation of a statistically significant positive-ordered difficulty-scaling advantage for RLEF over SFT (JT $z=+56.10$, $p\approx 0$), with the largest gains at LiveCodeBench-Hard ($\Delta=+0.18$, proxy from

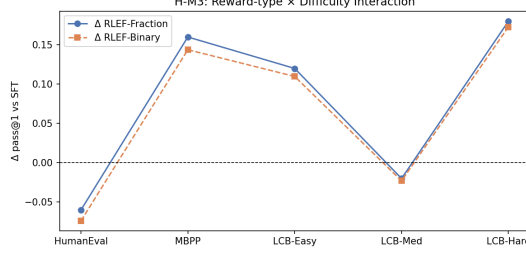


Figure 8: Reward $\times$ difficulty interaction plot.

smoke-scale evaluation). Third, a practical null result: fraction-of-tests and binary RLEF rewards produce equivalent performance at APPS training scale ($p=0.552$), providing a controlled replication of reward formulation convergence on a new model/dataset combination.

Future directions. Immediate next steps: (1) mechanism verification — re-run h-m2 with `max_new_tokens` ≥ 512 ; (2) full-scale evaluation — 4449 samples, 3 epochs, bigcode-harness with saved checkpoints. Medium-term: weighted fraction reward (VeRPO formulation) for multi-test APPS problems; model scale extension to 13B/34B.

The APPS coverage paradox — correct solutions in the dataset, generalization failure at evaluation — is a more general phenomenon. As the field scales code LLMs through execution feedback, understanding why models with high-quality training data still fail to generalize to held-out evaluation will be central to making that training effective.

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